EE-170L Computer Programming Lab

Lab Report No. 01 – Spring 2026

Lab Title: Installation of C++ on Microsoft Windows

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Student Name: Hidayat Ullah
Group Members:

- Hidayat Ullah
- Anis Ahmad
- Jawad Ali

Course: EE-170L Computer Programming Lab
Semester: Spring 2026

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1. Objective

The objective of this lab is:

- To install Visual Studio Code
- To install MinGW C++ compiler
- To configure environment variables
- To install required extensions
- To run a simple C++ program

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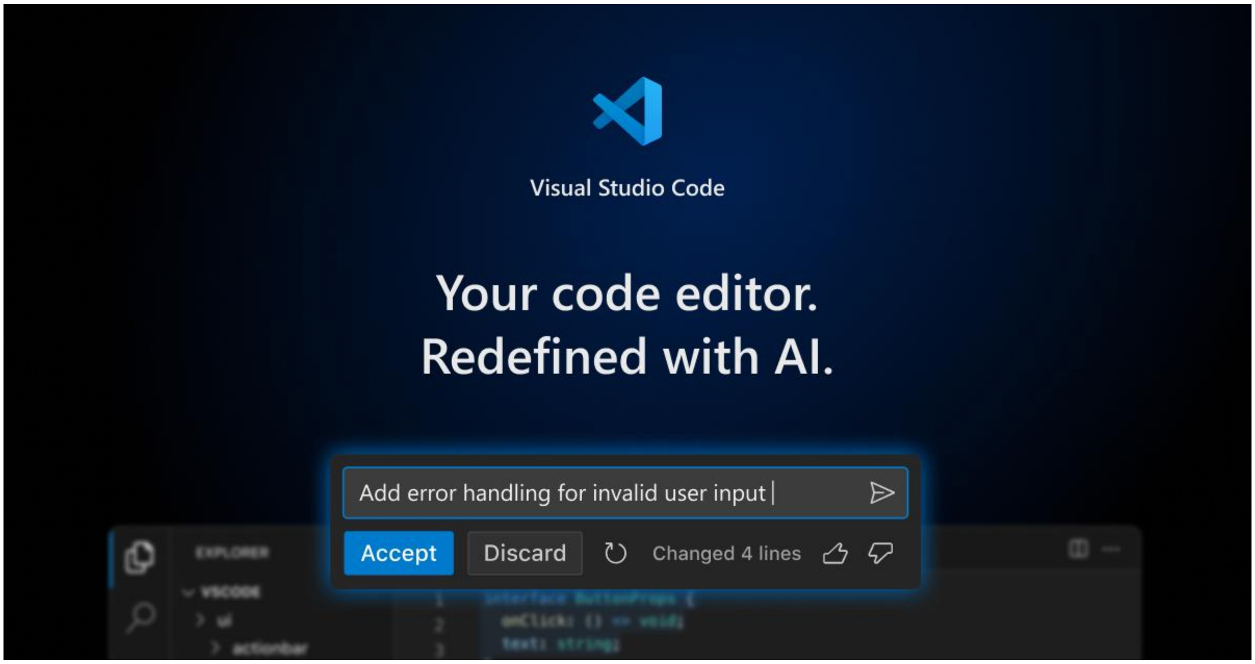
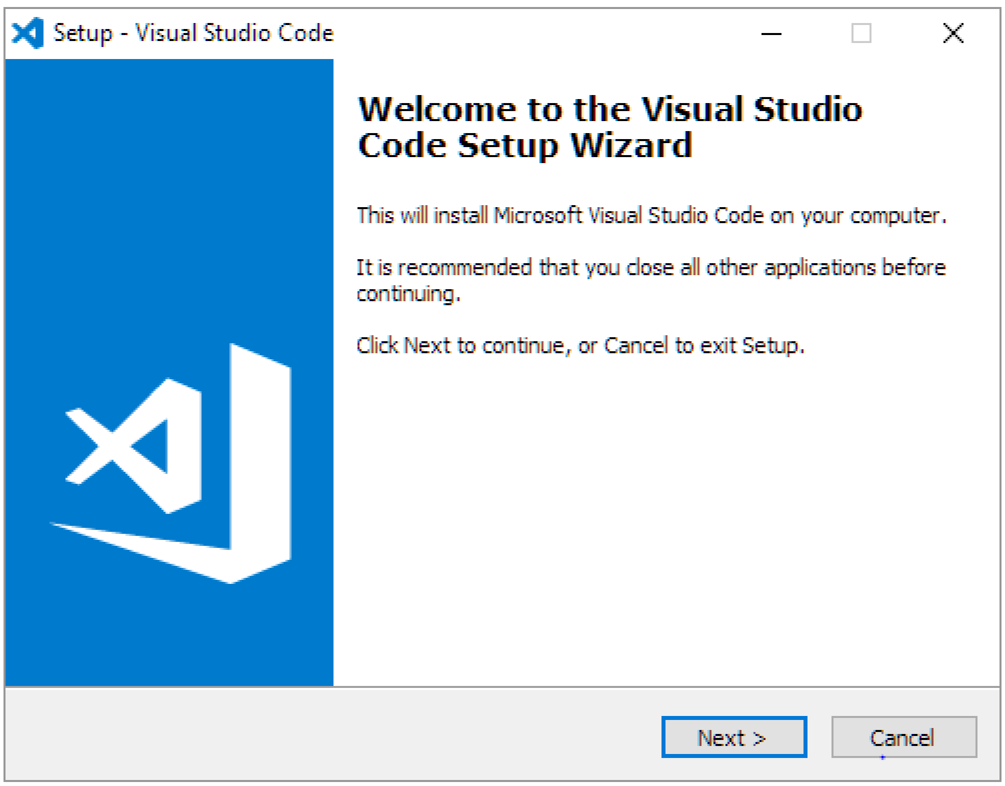
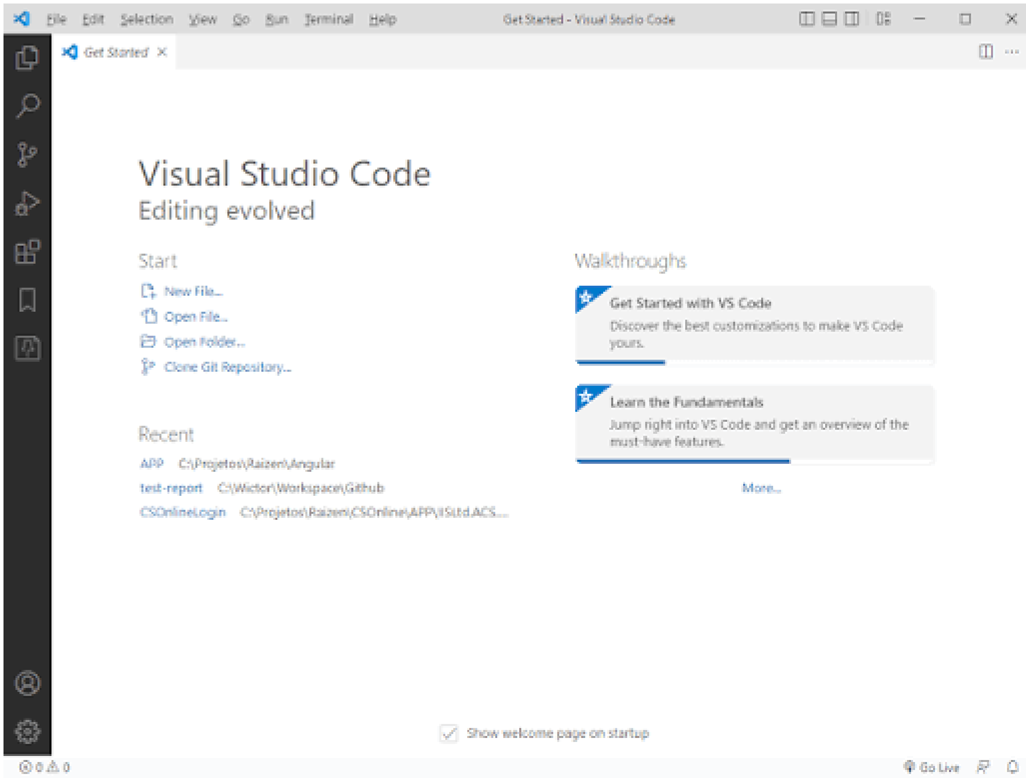
2. Software Installation

2.1 Installation of Visual Studio Code


Visual Studio Code
Your code editor.
Redefined with AI.
Add error handling for invalid user input
Accept
Discard
Changed 4 lines


Setup - Visual Studio Code
Welcome to the Visual Studio Code Setup Wizard
This will install Microsoft Visual Studio Code on your computer.
It is recommended that you close all other applications before continuing.
Click Next to continue, or Cancel to exit Setup.
Next >
Cancel



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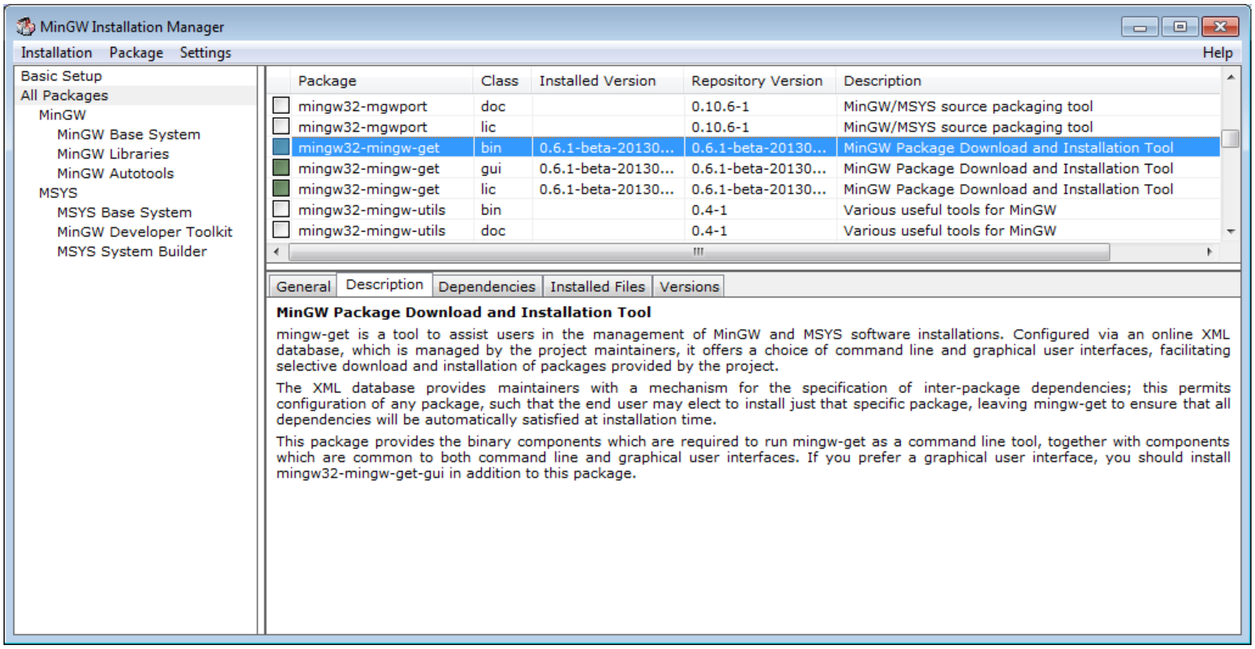
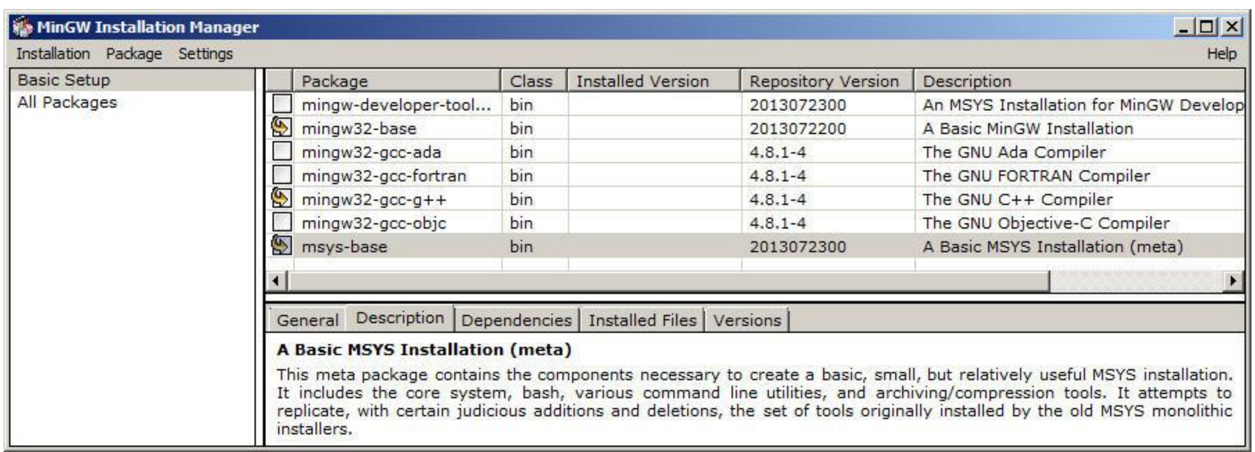
Steps:

1. Opened website https://code.visualstudio.com
2. Clicked on **Download for Windows**
3. Ran the installer
4. Followed installation steps
5. Visual Studio Code installed successfully

Purpose:
Visual Studio Code is used to write and run C++ programs.

2.2 Installation of MinGW Compiler





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Steps:

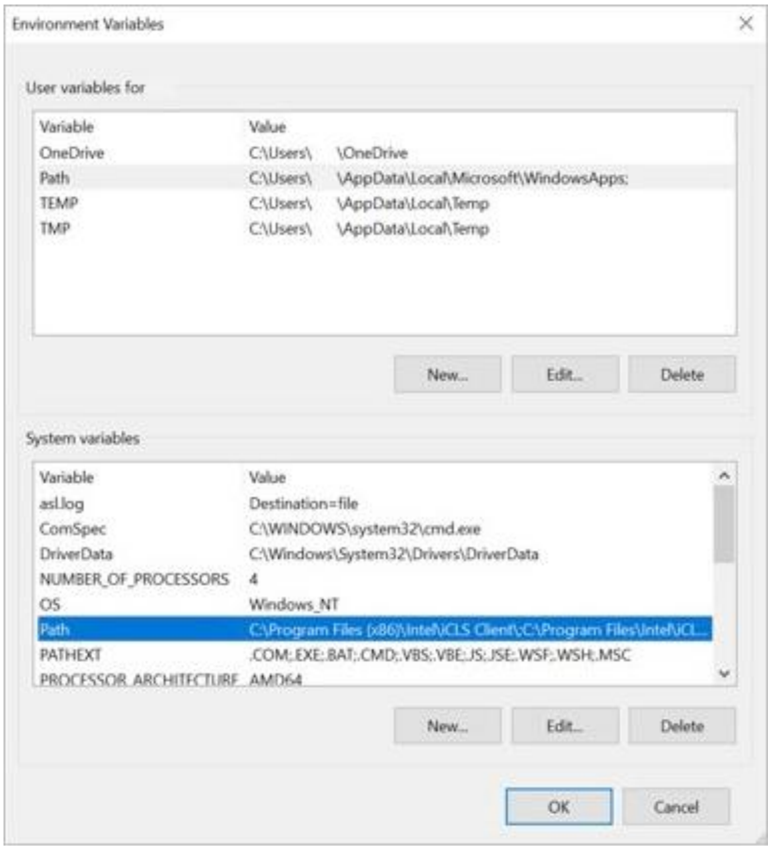
1. Opened website https://sourceforge.net/projects/mingw/
2. Downloaded MinGW installer
3. Installed MinGW software
4. Opened Installation Manager
5. Selected package:

mingw32-gcc-g++

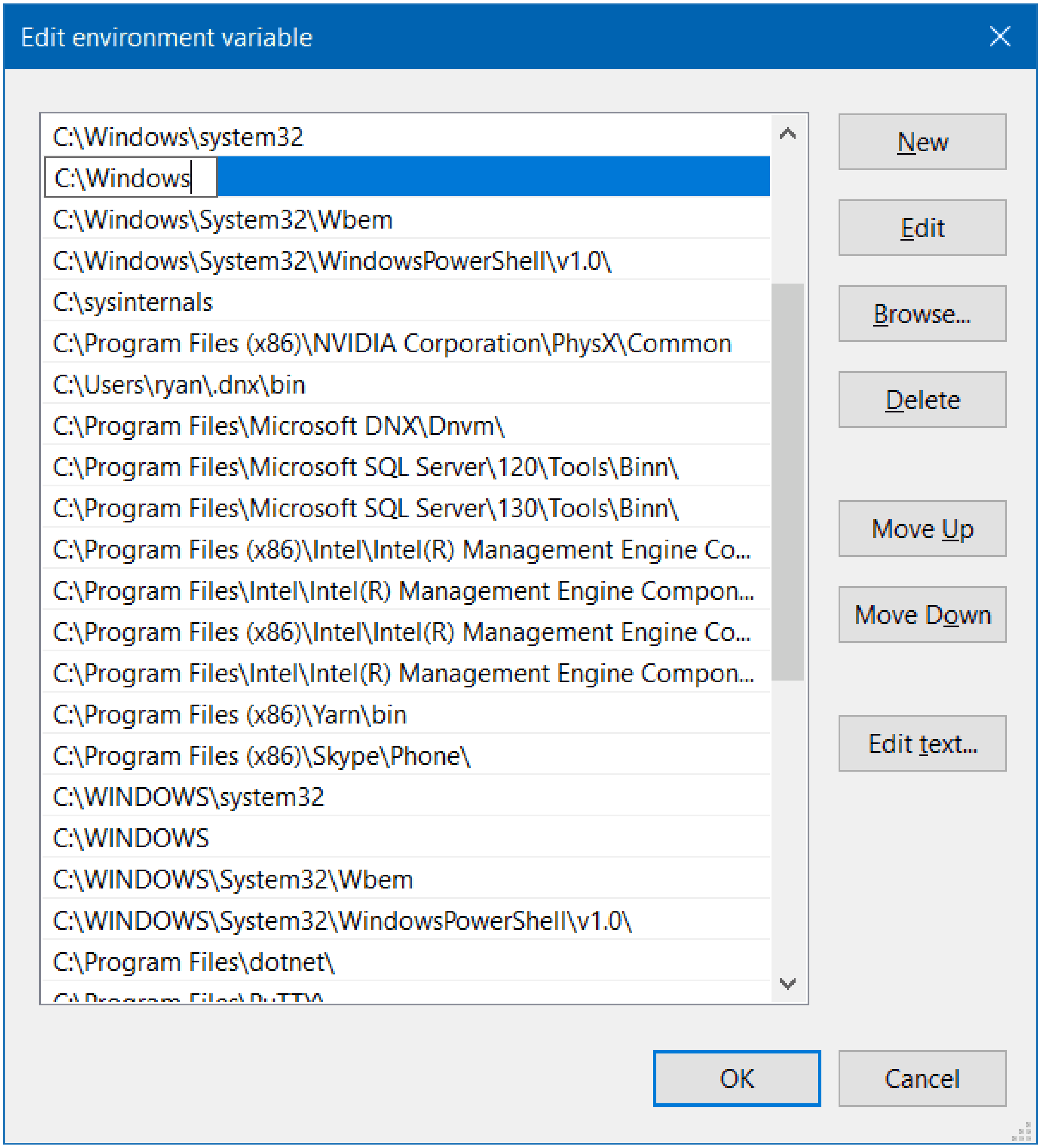
6. Clicked Install → Apply Changes

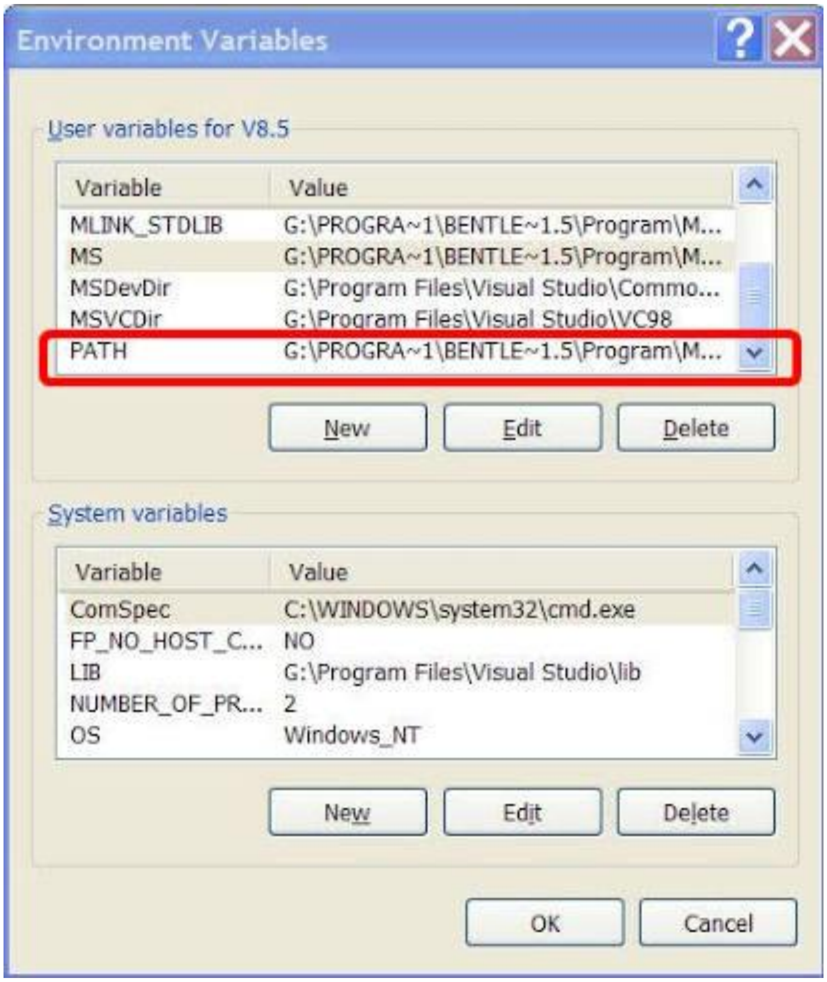
Purpose:
MinGW is used to compile C++ programs.

2.3 Adding MinGW Path to Environment Variables



ComputerHope.com


Edit environment variable
C:\Windows\system32
C:\Windows
C:\Windows\System32\Wbem
C:\Windows\System32\WindowsPowerShell\v1.0\
C:\sysinternals
C:\Program Files (x86)\NVIDIA Corporation\PhysX\Common
C:\Users\ryan\.dnx\bin
C:\Program Files\Microsoft DNX\Dnvm\
C:\Program Files\Microsoft SQL Server\120\Tools\Binn\
C:\Program Files\Microsoft SQL Server\130\Tools\Binn\
C:\Program Files (x86)\Intel\Intel(R) Management Engine Co...
C:\Program Files\Intel\Intel(R) Management Engine Compon...
C:\Program Files (x86)\Intel\Intel(R) Management Engine Co...
C:\Program Files\Intel\Intel(R) Management Engine Compon...
C:\Program Files (x86)\Yarn\bin
C:\Program Files (x86)\Skype\Phone\
C:\WINDOWS\system32
C:\WINDOWS
C:\WINDOWS\System32\Wbem
C:\WINDOWS\System32\WindowsPowerShell\v1.0\
C:\Program Files\dotnet\
New
Edit
Browse...
Delete
Move Up
Move Down
Edit text...
OK
Cancel



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Steps:

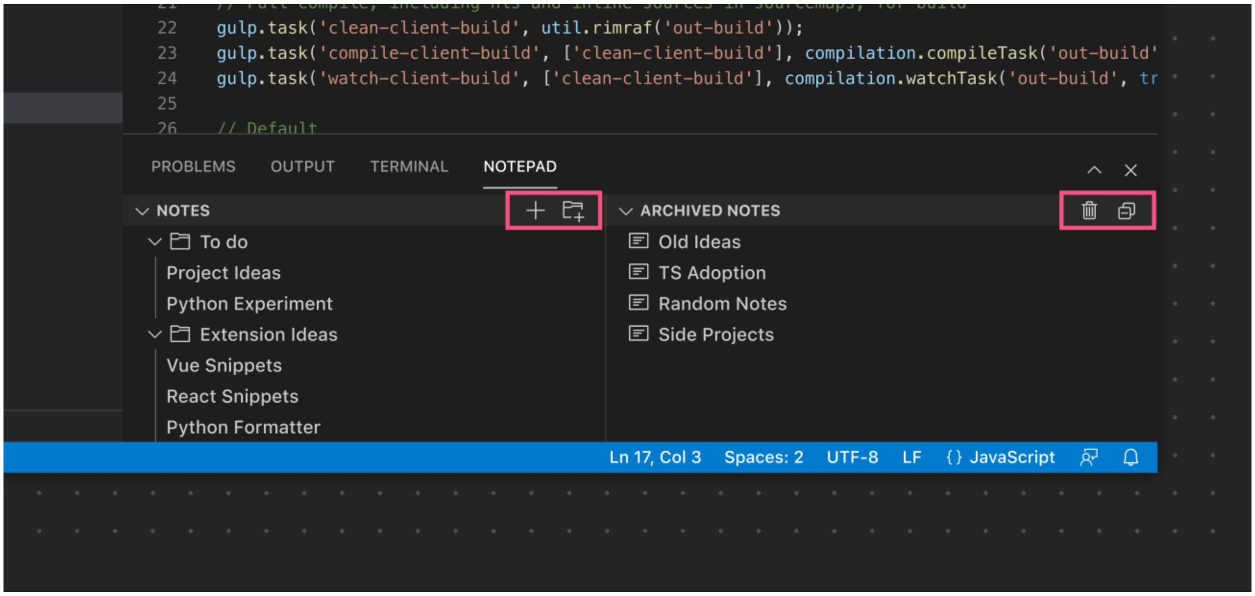
1. Copied path:

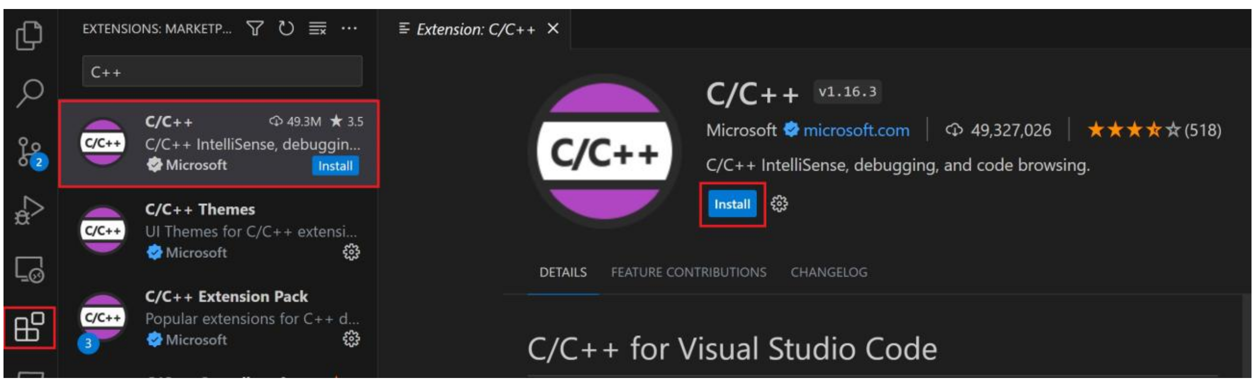
C:\MinGW\bin

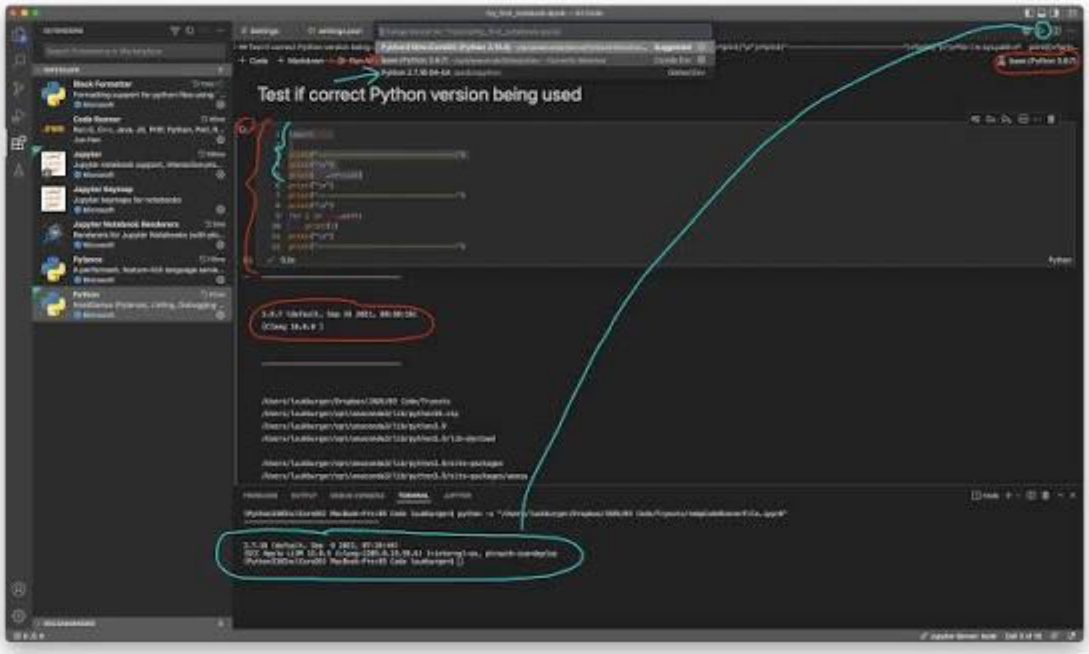
2. Opened Environment Variables
3. Clicked Path → Edit
4. Clicked New
5. Added path
6. Clicked OK

Purpose:
This allows system to recognize the g++ compiler.

2.4 Installation of Extensions in Visual Studio Code







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Extensions Installed:

- C/C++ Extension
- Code Runner Extension

Purpose:
These extensions help in compiling and running C++ programs.

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3. Lab Tasks

Task 1: Make group of three students

Group Members:

1. Hidayat Ullah
2. Anis Ahmad
3. Jawad Ali

Task 2: Understand lab instructions

All instructions were read and understood.

Task 3: Complete installation and setup

Successfully installed:

- Visual Studio Code
- MinGW Compiler
- Environment Variables
- Extensions

Task 4: Run a simple C++ program

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4. Program Code

```
#include<iostream>
using namespace std;

int main()
{
  cout<<"Group Members:"<<endl;
  cout<<"1. Hidayat Ullah"<<endl;
  cout<<"2. Anis Ahmad"<<endl;
  cout<<"3. Jawad Ali"<<endl;
  return 0;
}
```

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5. Output

Group Members:
1. Hidayat Ullah
2. Anis Ahmad
3. Jawad Ali

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6. Result

Visual Studio Code and MinGW compiler were installed successfully.
The program compiled and executed without errors.

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7. Conclusion

In this lab, we learned:

- Installation of Visual Studio Code
- Installation of MinGW Compiler
- Setting Environment Variables
- Installing Extensions
- Running C++ Program

The C++ development environment is now ready for programming.